

The Human Body

Skeletal System



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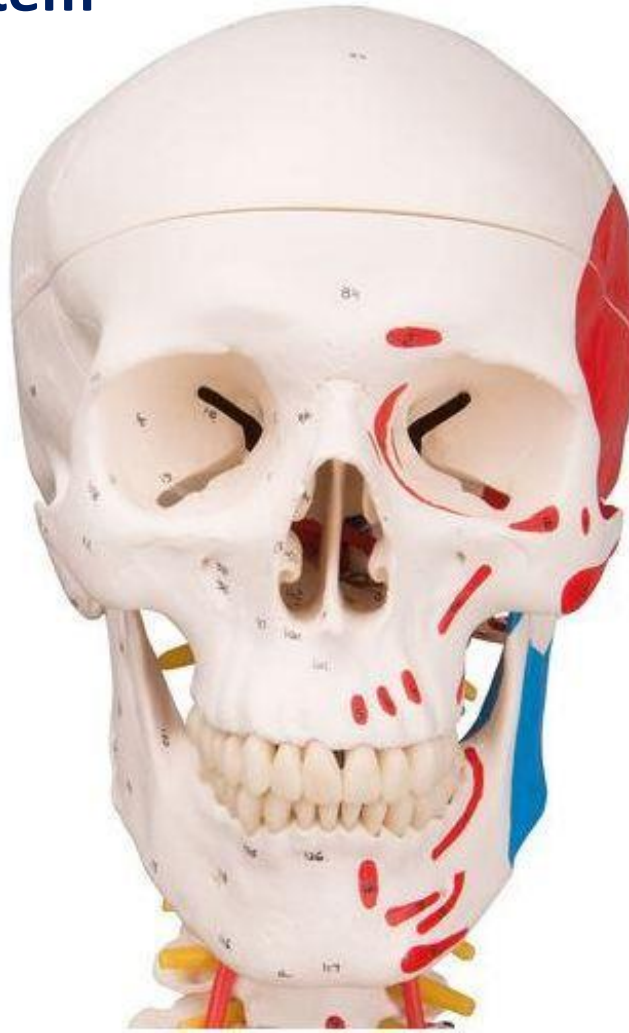
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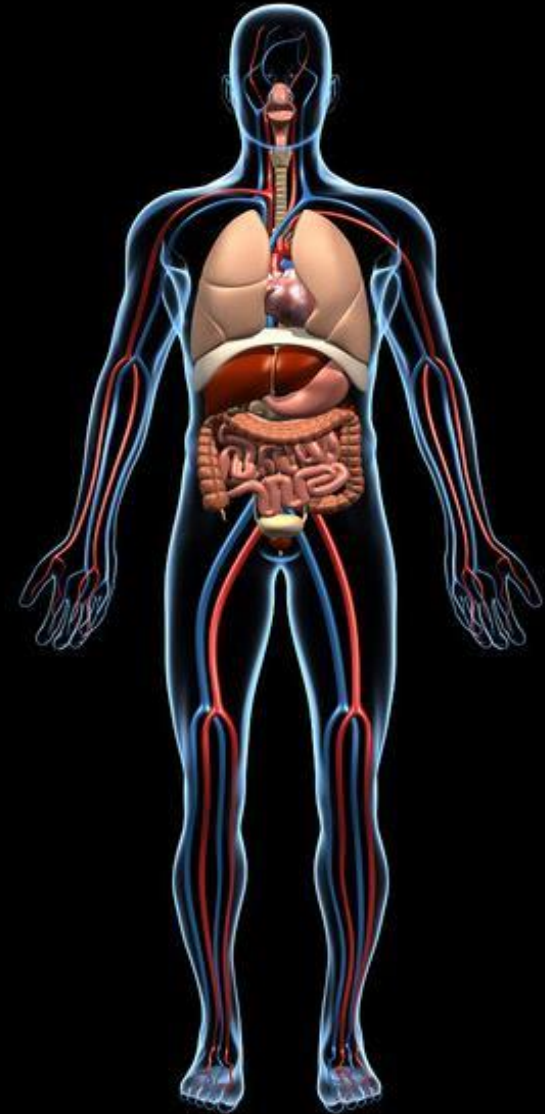
The Human Body

The human body is a wonderfully efficient and complex machine. It has organs and systems that are specialized to perform specific functions.

The stomach, the intestines, the liver and the pancreas help with the digestion of food. These organs make the Digestive system.

Lungs help in breathing. The lungs are one of the most important organs of the Respiratory system.

The system that helps you feel and react is the Nervous system. The system that helps in movement is the Locomotory system.





Can you feel something hard below your skin?

These are the bones.

Have you ever wondered what your body would be like if there were no bones in it?

Let us explore the main functions of the bones.

A human adult has 206 bones, whereas a human child has more than 350 bones. As a child grows, some of the bones fuse together.



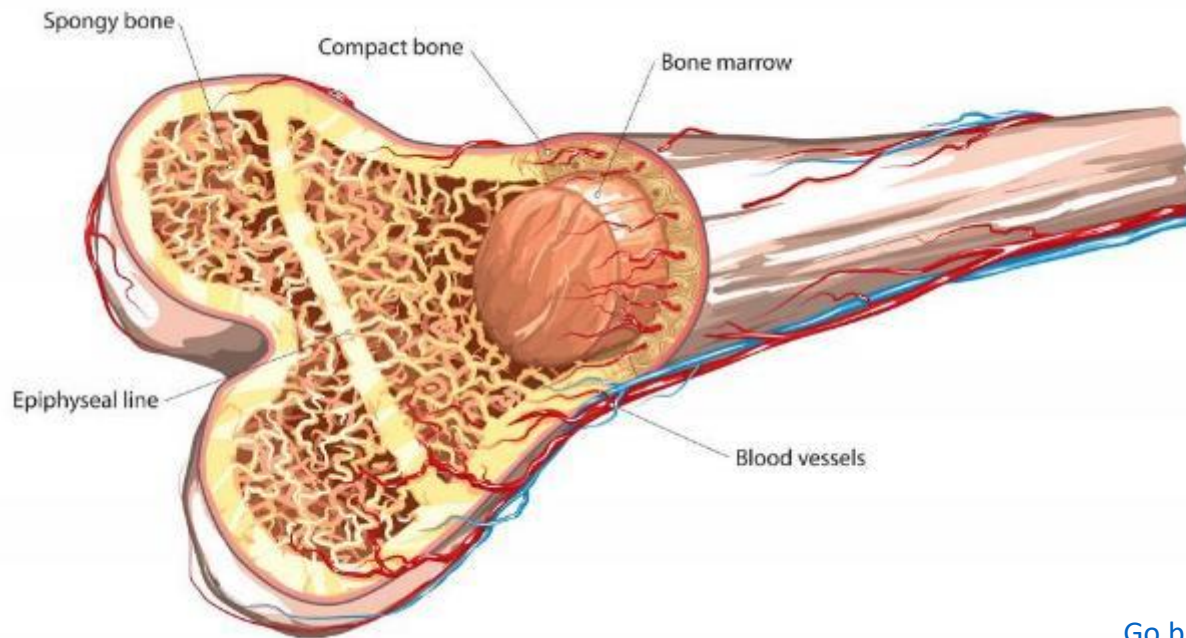
The bones of the body are living and need nourishment for their growth and repair. Minerals like calcium, phosphorous and certain vitamins are important for bones.

Bones differ in size and shape. The bones of your fingers and toes are small, but the bones of your arms and legs are long.

BONES

Bones have a supply of blood vessels and nerves which grow as you grow older. Bones are hollow and not solid. The inside of bones contain a soft and spongy matter called **bone marrow**. It has blood vessels.

Bone marrow is the body's factory for producing red blood cells and it also stores fat.





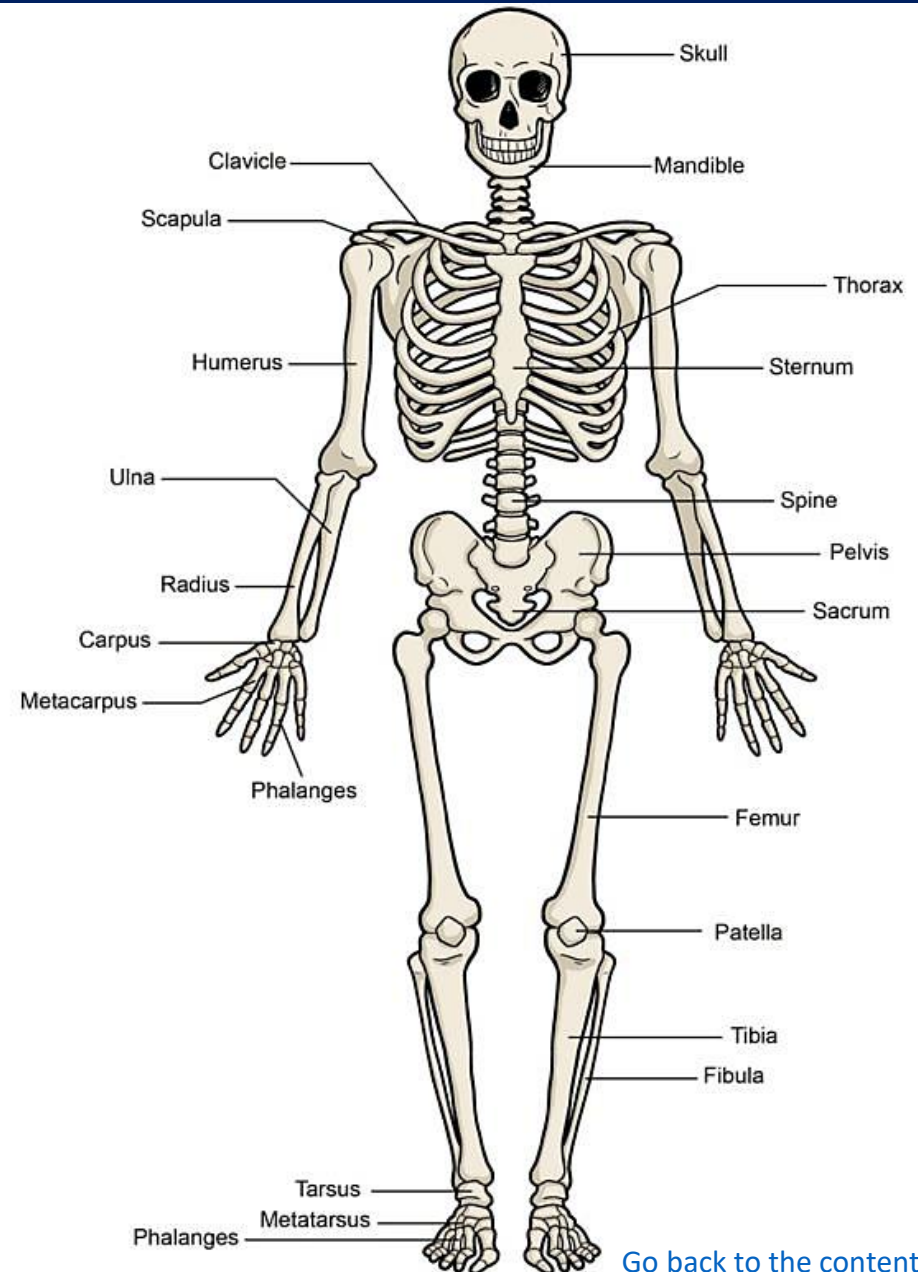
The skull is a bony structure in our body that **protects the brain**. In adult human beings, the skull is normally made up of 22 bones, joined together to cover the brain and give shape to it.

All the bones in the skull are fixed, except the lower jaw, which is movable. The movable jaw allows us to talk and eat.

THE SKELETON

The skeleton is the framework of bones which gives shape to the body and protects the soft, delicate inner organs such as the heart and lungs.

The skeleton also enables the movement of the body.



Wizard's Corner

The study of bones is known as osteology.

THE RIBCAGE

The **ribs** are narrow curved bones that form a cage to **protect** the **heart** and **lungs**. There are 12 pairs of ribs in a human being. A long bone at the centre of the chest, called the **sternum**, holds the ribs in place in the front. The ribs are also attached to the backbone.

However, the last two pairs of ribs are attached to our backbone and not the sternum. They are called **floating ribs**.



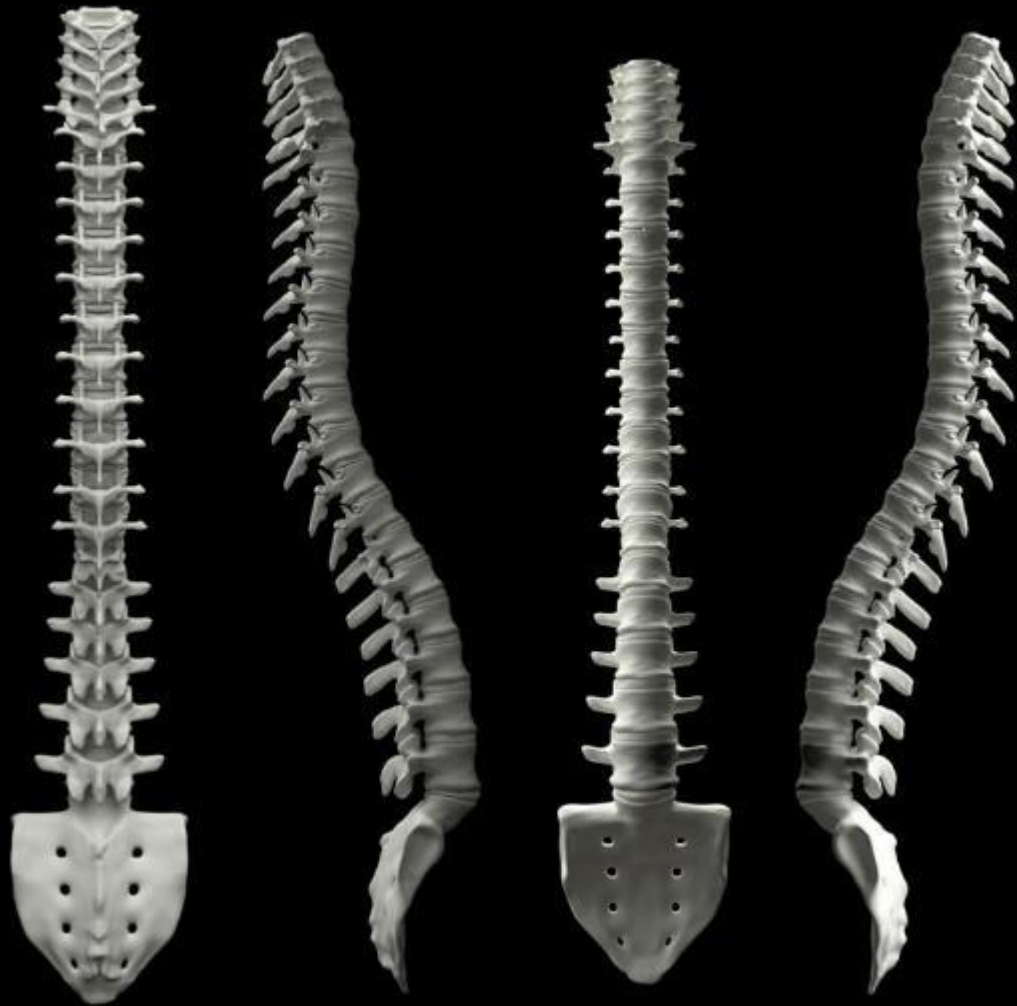
Wizard's Corner

Animals such as tigers and snakes have a backbone and are called vertebrates. Animals such as earthworms and cockroaches do not have a backbone and are called invertebrates.

THE BACKBONE / SPINE

The backbone, known as the **spine** or the **vertebral column**, is attached to the skull. It is made up of 33 small bones. Each bone is called a **vertebra**. The vertebrae can move over each other. This allows us to bend and twist our back.

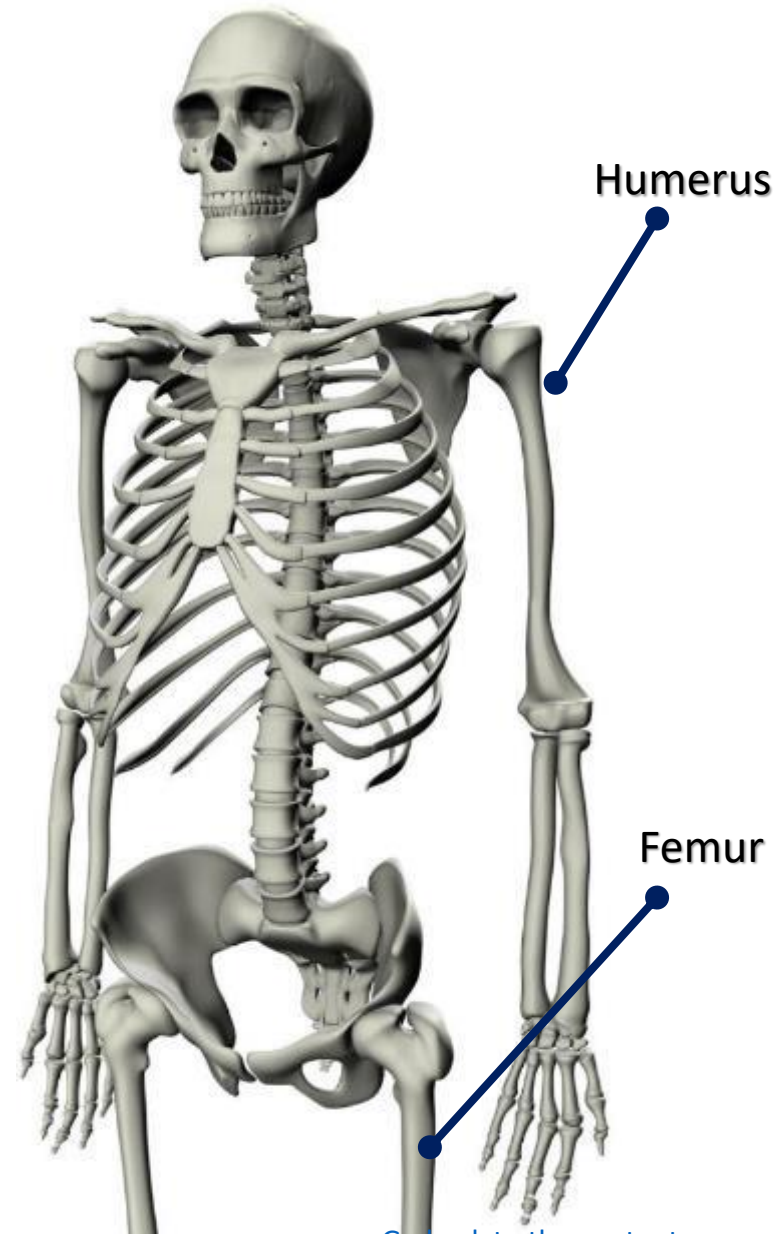
The spine is the central supporting rod for the skeleton and protects the **spinal cord**, which is a thick bundle of nerves.



1 The arms and legs are our limbs. The bones of our limbs are very long.

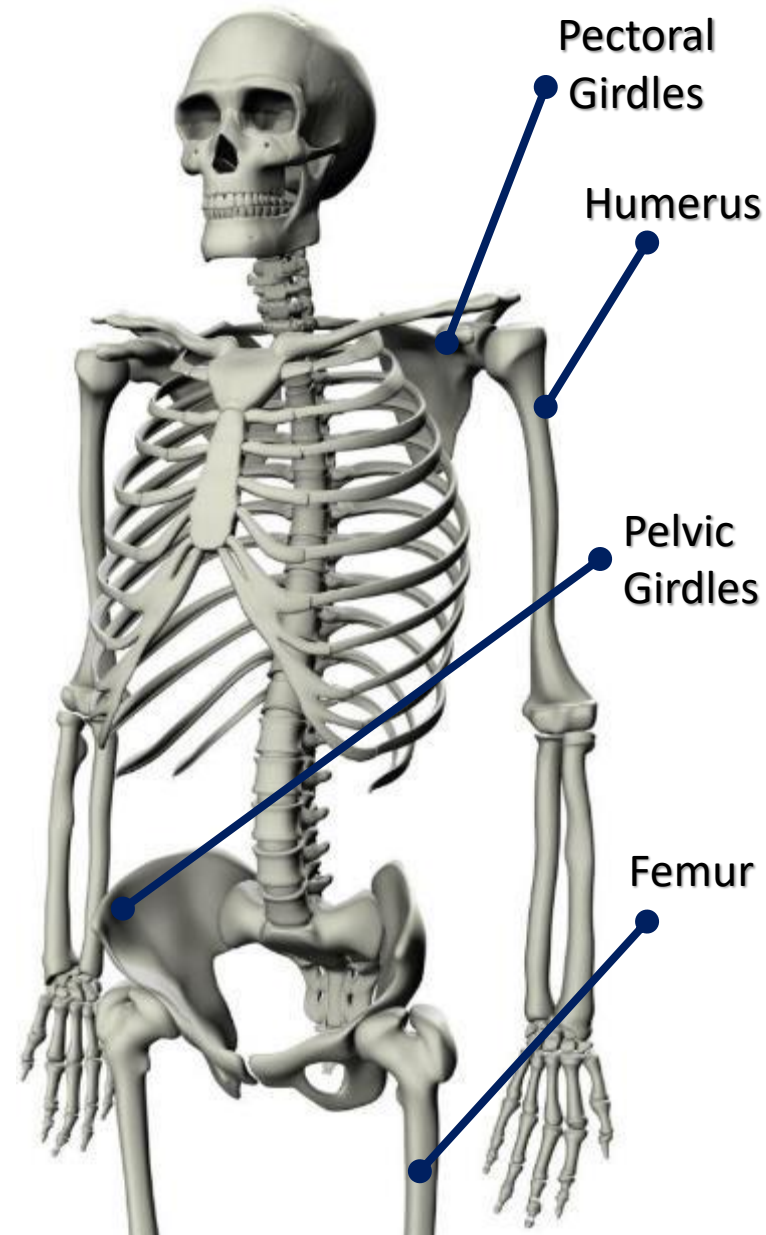
2 The upper arm has one long bone called **humerus**. The lower arm has two bones which are called the **ulna** and the **radius**. The limbs are not joined directly to the backbone, but they are joined through **girdles**.

Girdles are the bony structures to which the limbs of vertebrates are attached.



3 The upper arm is joined to the spine by the **collar bone** and **shoulder girdle** called the **pectoral girdle**. The hand and the wrist have many small bones.

4 The thigh has a long bone. It is the longest bone in our body and is called **femur**. The leg has two bones. The thighs are joined to the spine by the **hip girdle** called **pelvic girdle**. Many small bones make up the ankle and the foot.



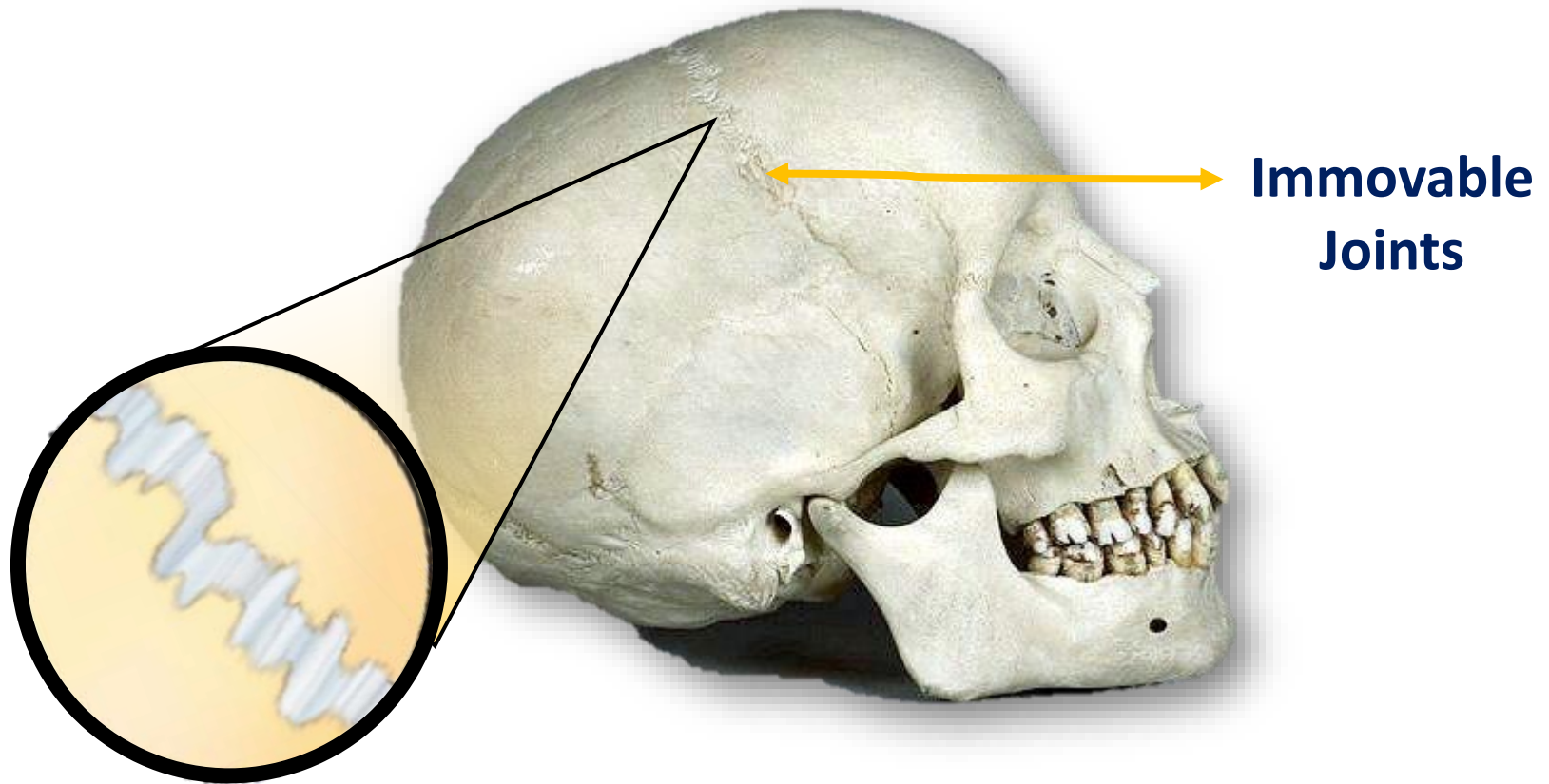
There are two types of joints – **immovable** and **movable**.



Joints help with the **movement** of the body.

IMMOVABLE JOINTS

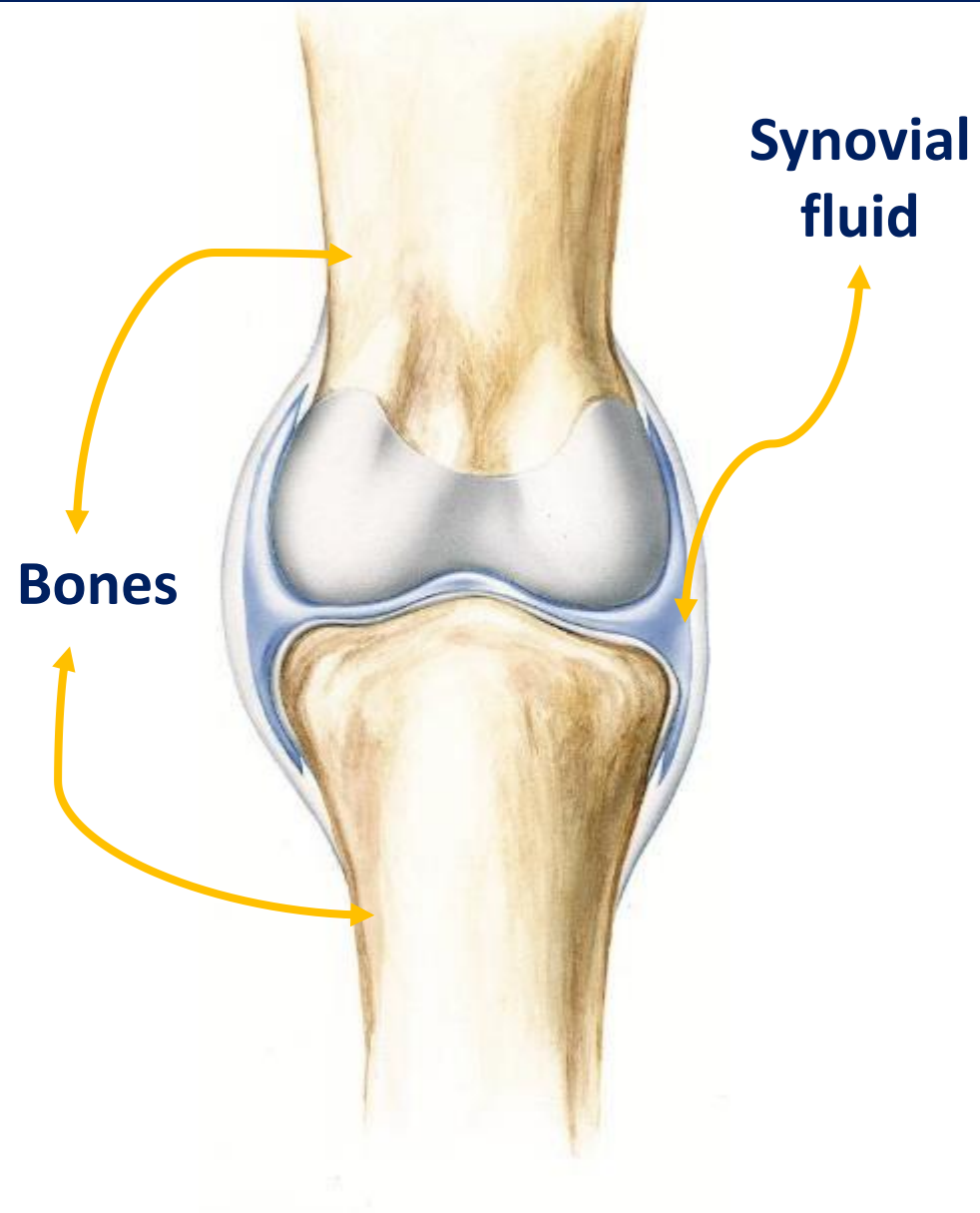
At these joints, movement in between the two bones does not occur as the bones are firmly attached to one another. This type of **joint** is present in between the bones of the **skull**.



MOVABLE JOINTS

These joints provide free movement of the body.

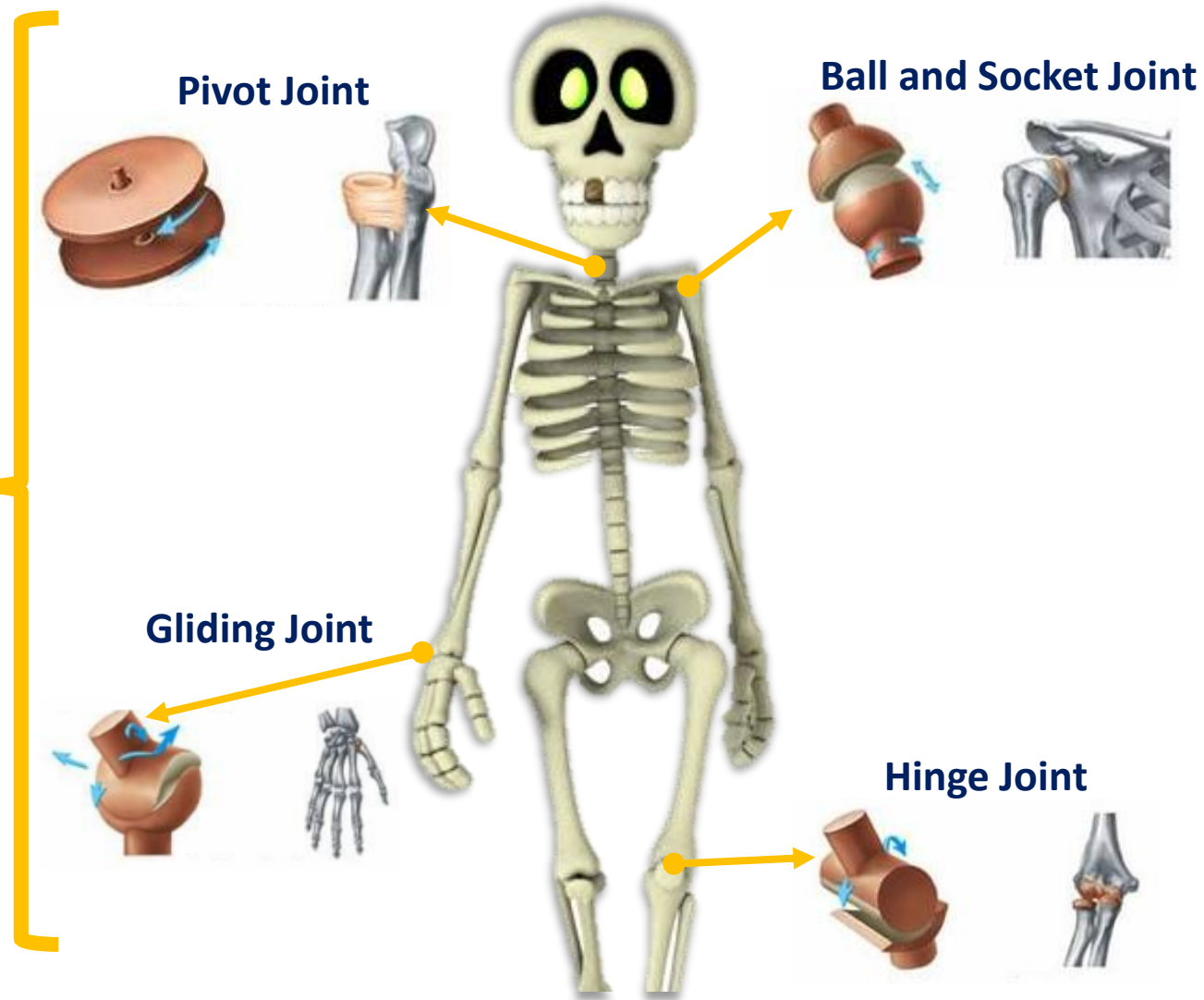
This is because of a fluid known as **synovial fluid**, present between the bones, which acts like a lubricant. The synovial fluid reduces friction between the bones of the joint so that they can move easily.



MOVABLE JOINTS

The different kinds of movable joints in the body are:

- Hinge joints
- Ball and Socket joints
- Pivot joints
- Gliding joints



HINGE JOINTS

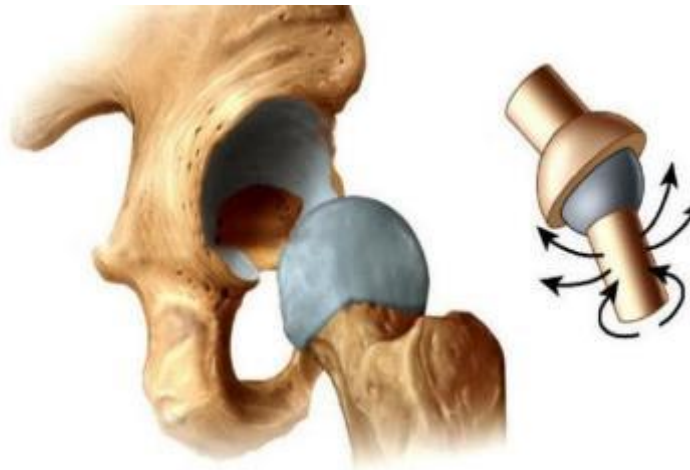
Hinge joints are like the hinges in a door. They enable movement similar to the opening and closing of a hinged door. This type of joint is found in knees, elbows, fingers and toes.



BALL AND SOCKET JOINTS

In a ball and socket joint, one end of the bone is like a ball that fits into a hole or a socket of the other bone. Such a joint allows free movement in all directions.

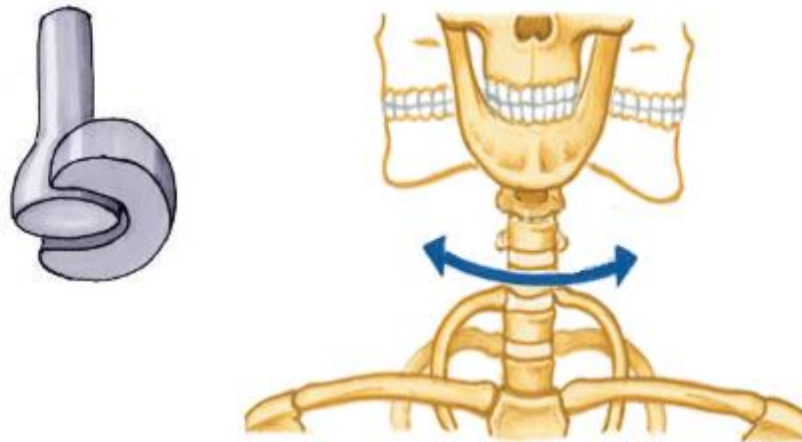
The shoulder joints and hip joints are examples of ball and socket joints. They allow you to swing your arms and legs in many different directions.



PIVOTAL JOINTS

The pivotal joint is found between the first two vertebrae of the backbone.

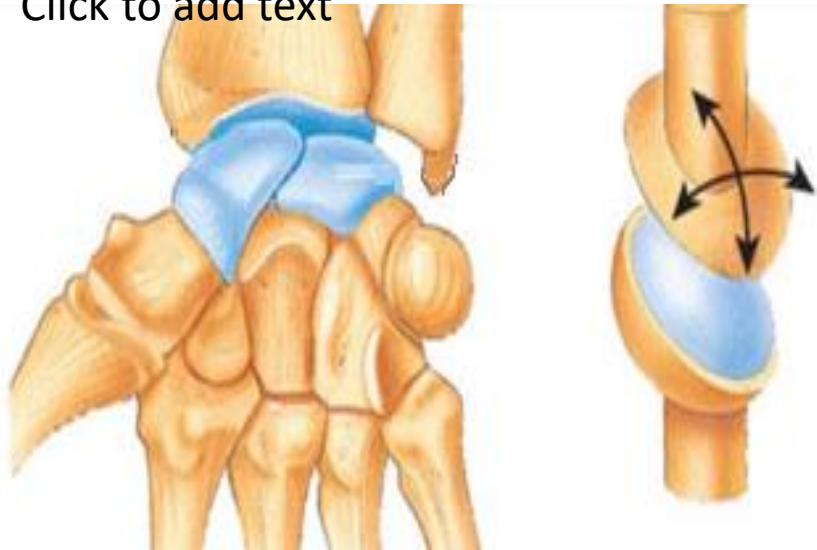
The skull is connected to the top vertebra with a pivotal joint. It allows the head to move upwards, downwards and sideways.



GLIDING JOINTS

Gliding joints allow the bones to glide past one another in any direction along the plane of the joint — up and down, left and right, and diagonally.

Click to add text



Wizard's Corner

Arthritis is a disease of the joints. It causes inflammation, pain, swelling and stiffness.

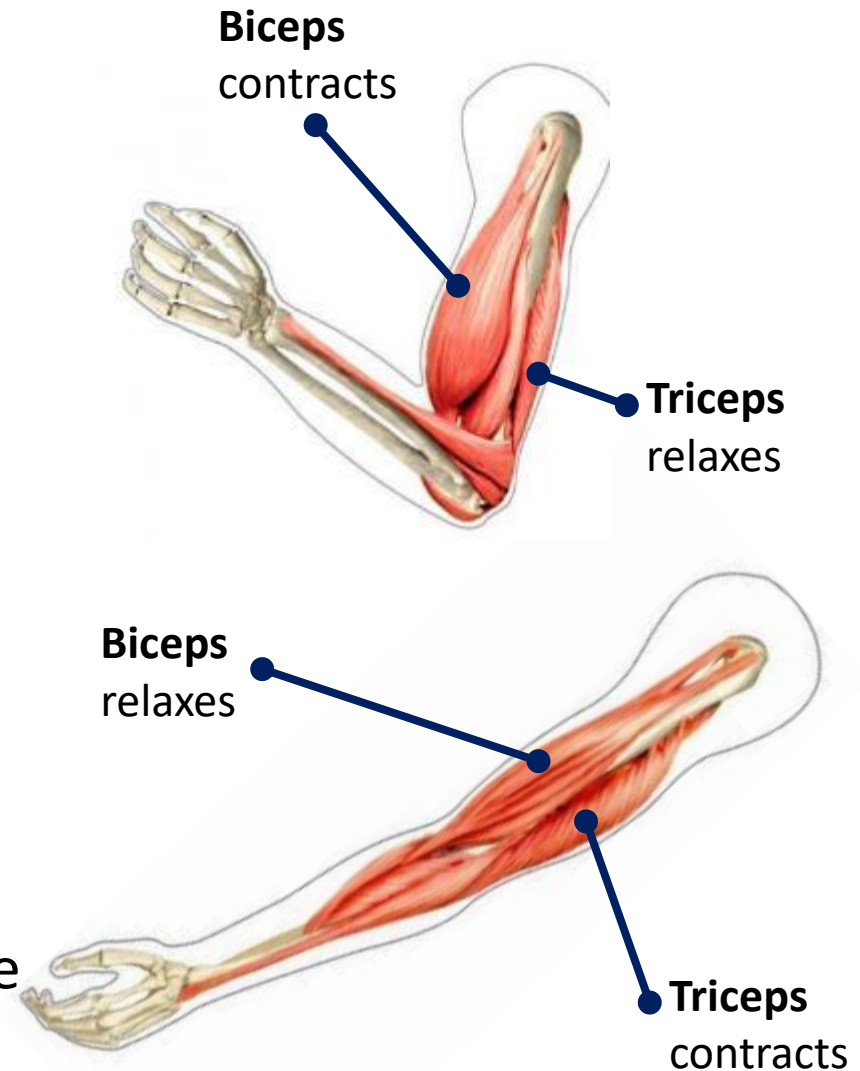


Muscles lie under the skin. They are present in the heart, digestive organs, and blood vessels. They make our heartbeat. They help us move food in the food pipe and help us chew our food. Muscles are made up of a bundle of fibers.

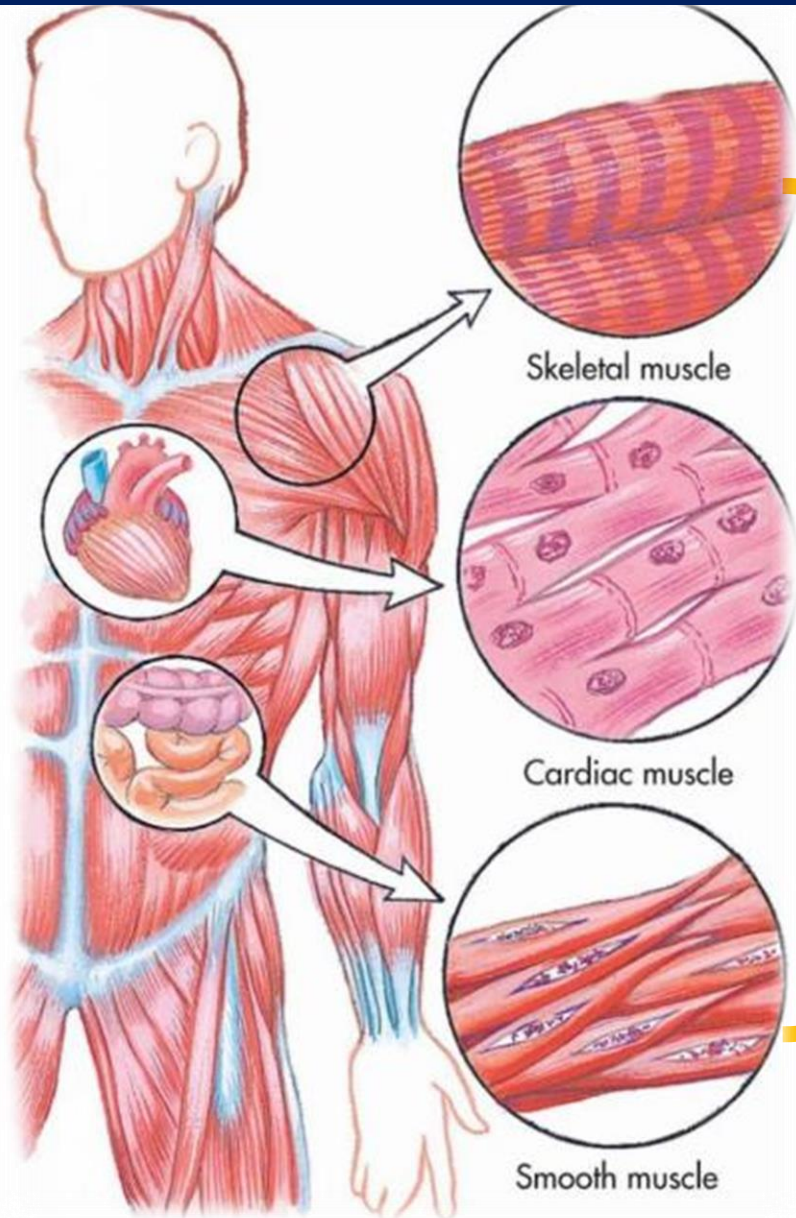
Muscles not only help the body parts to move but also give them strength to do work. They are attached to the bones by strong fibers called **tendons**.

How Do MUSCLES WORK?

- 1 Muscles work by pulling or contracting. Muscles cannot push. Two sets of muscles work on a joint to bring about the movement.
- 2 The muscles of the upper arm are called **biceps** and **triceps**.
- 3 When the **biceps** contract, the **triceps** relax. This contraction and relaxation of the muscles make the bones move.



KINDS OF MUSCLES



There are three types of muscles in our body: voluntary, involuntary and cardiac.

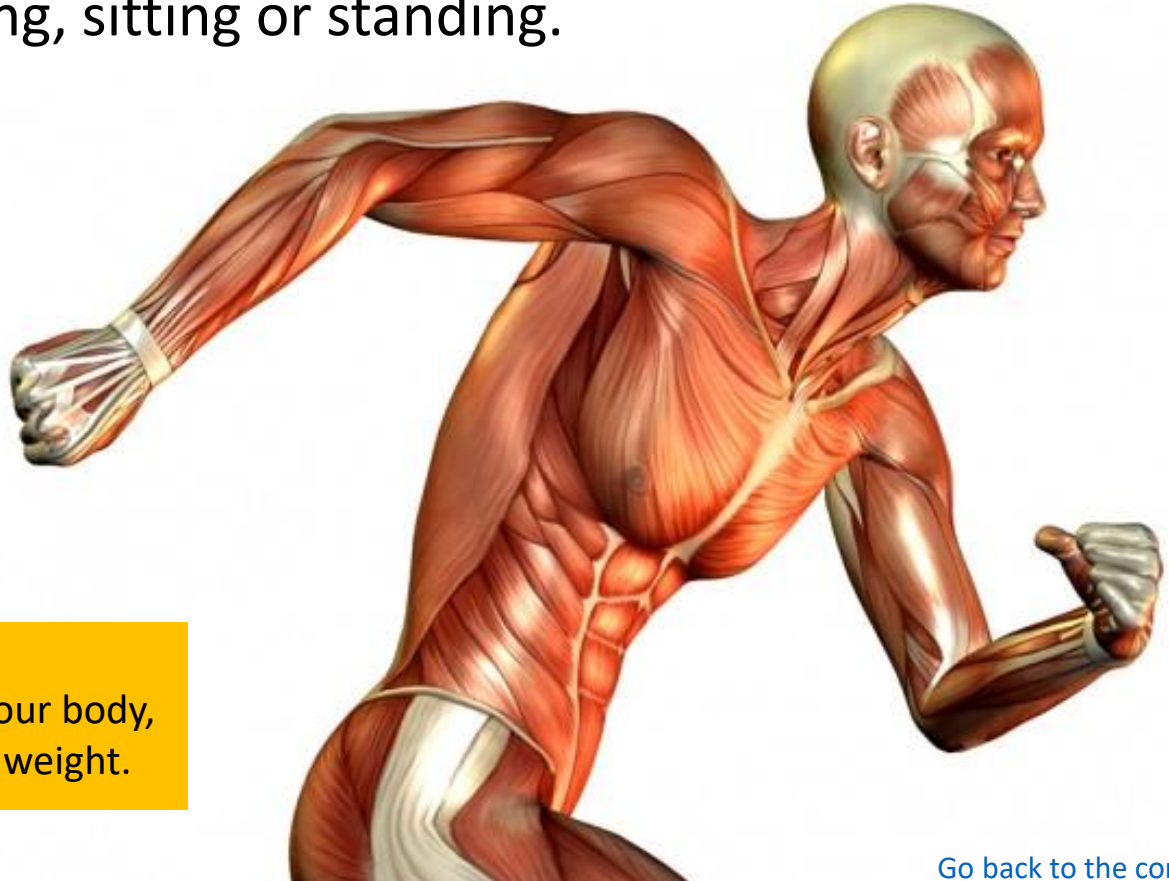
Voluntary or skeletal muscles are attached to our skeleton and are under our control. These muscles are found in the arms, legs and eyes.

Involuntary muscles or smooth muscles are not under our control. They work automatically. They are found in our stomach and intestine.

Cardiac or heart muscles are found in the heart. They are involuntary and work throughout our life without stopping or getting tired.

TO MAINTAIN STRONG MUSCLES

Exercise is a good thing to improve muscle strength. You may have observed that athletes and sportspersons have well-developed muscles. You should also play outdoor games to keep your muscles strong and in good shape. It is also important to maintain a correct posture while walking, sitting or standing.



Wizard's Corner

There are about 640 muscles in our body, which make up 40% of the body weight.

- 1 A skeleton is the framework of bones that gives shape, protects organs and provides support to the body
- 2 The skeleton consists of bones of different shapes and sizes.
- 3 The skull, spine, ribcage, limbs, shoulder girdle and hip girdle are parts of the skeleton.
- 4 Bones are hard on the outside, and inside they contain bone marrow, a soft and spongy matter, where red blood cells are formed.



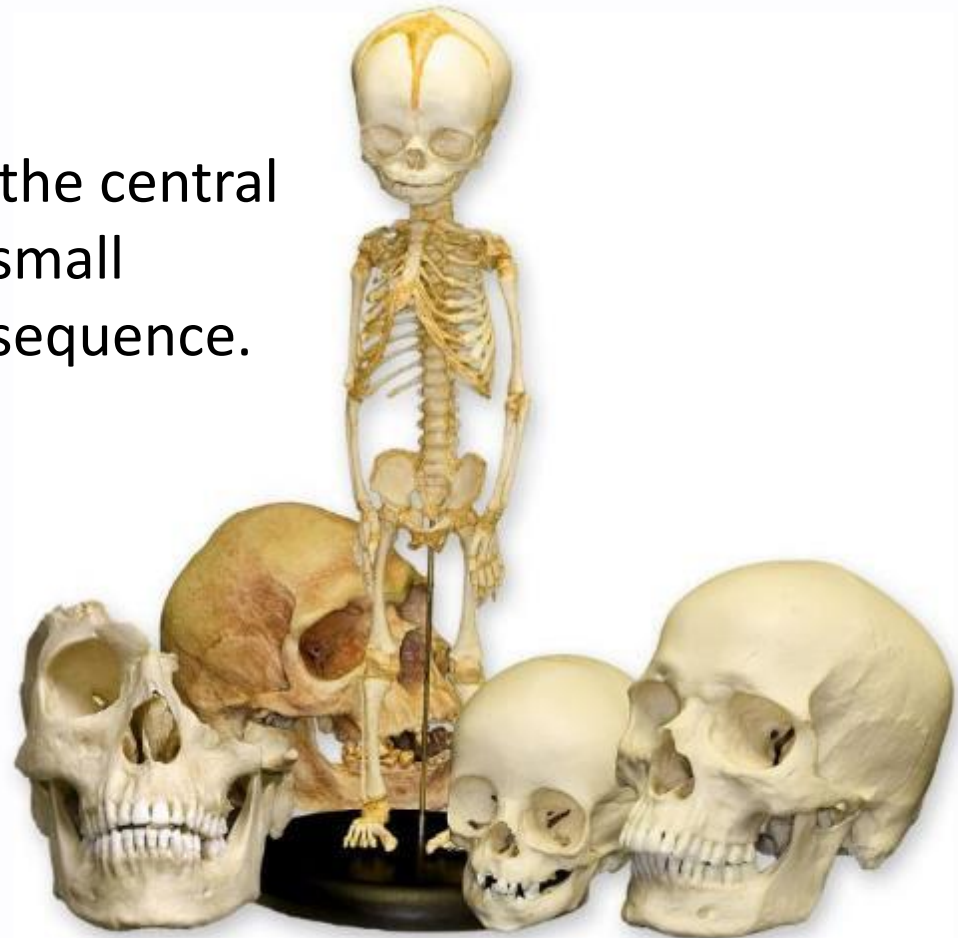
- 5 Joints help the bones to move. They can be movable or immovable.
- 6 There are 4 kinds of movable joints: hinge, gliding, pivot, and ball and socket.
- 7 Muscles help the bones to move by pulling at them. There are three kinds of muscles - voluntary, involuntary and cardiac.



Activity 1: Press your hands on your chest, you will feel the bones and not the lungs or the heart. In the same way try to feel bones in other parts of your body.

Activity 2: Run your fingers along the central portion of your back and feel the small bones of the spine, arranged in a sequence.

Activity 3: Move your arms, your legs, your wrists and your head. This way you can feel the working of different joints in your body.



1 Does it hurt you when you break a bone?
Or when you hit your arm against
something hard?
What does this show?

2 What would happen if our backbone was
made of a single long bone, instead of many
small bones?

3 Name the parts of your body, where you have the following
joints: the gliding, the pivot, the ball and socket, the hinge joint.



biceps: muscles present in the upper arm that help us to bend at the elbow

cardiac muscles: muscles present in the heart

involuntary muscles: muscles that are not under our control

joint: two or more bones meet to form a joint

ligament: connective tissue that joins one bone to another

tendon: connective tissue that joins a muscle to a bone

triceps: muscles present in the lower arm that help us to straighten the elbow

voluntary muscles: muscles that are under our control